Lesson 1 Connecting tree rings and radiocarbon

For slide 11 and Lesson 3, slide 13. A four-minute check; no preparation is required for Lesson 1.

Two clocks, connected by known-age wood

Crossdate rings to establish their calendar years. Measure their radiocarbon content. Many pairs of known calendar ages and measured radiocarbon ages provide evidence for a calibration curve. Tree rings are a major source of calibration data, but do not supply the entire curve.

For an unknown sample, use the relationship in the other direction: compare the radiocarbon measurement and its uncertainty with an appropriate curve to obtain calendar-age ranges and probabilities. Atmospheric radiocarbon has varied, so subtracting a radiocarbon age from 1950 does not by itself give a calendar year.

Question	Evidence needed
Calendar age of a ring	Crossdating against a securely dated reference
Calendar age of an unknown sample	Radiocarbon measurement, uncertainty and suitable curve
Date of an archaeological activity	Relationship between growth, use, deposition and the activity

Why measure radiocarbon in wood whose calendar age is already known? Explain the separate roles of crossdating and radiocarbon measurement.

Check

The independently known calendar age makes the ring a calibration reference. Irregularities in the curve can produce several possible calendar-age ranges for one measurement.

Historic England: radiocarbon calibration